Technical Design Report

Defence Drone Navigation using GBFS and A* Search
Course: AIMLCZG557 / AECLZG557
Assignment: PS4 – Defence Drone Navigation System
Group: G116

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1. Project Summary

This project develops an intelligent Defence Drone Agent capable of navigating an 8×8 restricted airspace to locate an external signal source. The drone starts at (0,0) and must reach the target at (6,7) while avoiding No-Fly Zones and minimizing exposure to Weather Hazard regions. The solution implements:

Greedy Best-First Search (GBFS) using Euclidean Distance (h1)
Greedy Best-First Search (GBFS) using Bounding-Box Risk-Weighted Heuristic (h2)
A* Search using Euclidean Distance

The objective is to evaluate search efficiency, path quality, computational complexity, memory usage, and heuristic effectiveness.

2. Problem Modelling and System Design

Environment Representation

The airspace is represented as an 8×8 grid loaded from the input file.

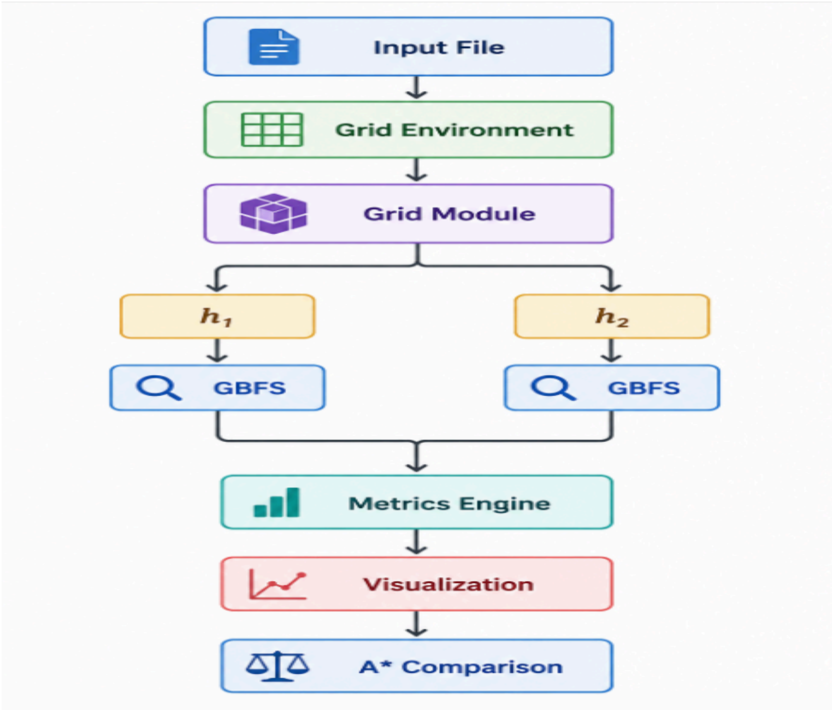
Symbol	Description	Cost
S	Start Node	1
E	Goal Node	1
.	Safe Airspace	1
W	Weather Hazard	4
N	No-Fly Zone	Blocked

Environment Statistics:

- Grid Size = 8 × 8
- No-Fly Zones = 8
- Weather Hazards = 7
- Start = (0,0)
- Goal = (6,7)

The generated environment visualization confirms obstacle placement and traversal constraints.

System Architecture & Design Flow:



Design Decisions:

Component	Design Choice	Reason
Environment	2D Matrix	Fast access O(1)
OPEN List	Priority Queue (Heap)	Efficient node selection
CLOSED List	Hash Set	Prevents revisits
Parent Tracking	Dictionary	Fast path reconstruction
Search Strategy	GBFS	Heuristic-guided navigation
Benchmark Algorithm	A*	Optimal path comparison

The implementation follows a modular design consisting of Grid, Heuristic, Search, Visualization, and Analysis modules

3. Heuristic Engineering and Search Strategy

Heuristic h1 – Euclidean Distance

The Euclidean heuristic guides the drone toward the goal using the shortest straight-line distance.
Features: Fast and computationally efficient and Goal-oriented navigation.

Heuristic h2 – Bounding-Box Risk-Weighted

This heuristic combines distance with environmental risk by evaluating the average cost of cells between the current node and the goal.
Features: Future-aware navigation and considers hazards and obstacles.

Search Strategy

GBFS selects the node with the minimum heuristic value and expands it until the goal is reached.

1. Select lowest heuristic node
2. Expand valid neighbours
3. Update OPEN and CLOSED lists
4. Repeat until Goal is reached

Data Structures Used:

Structure	Purpose
Priority Queue	OPEN list
Hash Set	CLOSED list
Dictionary	Parent tracking
Grid Matrix	Environment

Node Priority: North → East → South → West

Outcome:
h1 provides faster convergence, while h2 provides safer and more informed navigation. A* is used as a benchmark for optimal path comparison.

4. Results, Analysis and Comparison

GBFS Results

Both heuristics successfully reached the target node **(6,7)** while avoiding all No-Fly Zones. Each solution generated a path of **13 moves** and crossed **one Weather Hazard Zone**, resulting in a total path cost of **17**.

Heuristic Behaviour

- **h1 (Euclidean Distance)** followed the shortest geometric route toward the goal.
- **h2 (Bounding-Box Risk-Weighted)** selected a slightly different route by future environmental risk.
- Both heuristics reached the goal efficiently without entering dead-end regions.

Performance Comparison:

Metric	GBFS h1	GBFS h2	A*
Nodes Expanded	14	14	29
Runtime (ms)	0.13	0.19	0.21
Memory Usage	23	28	40
Path Cost	17	17	14

Analysis:

- **GBFS h1** achieved the fastest execution due to its simple distance-based evaluation.
- **GBFS h2** provided more informed navigation by incorporating environmental risk.
- **A*** expanded more nodes and consumed more memory but produced the lowest-cost path.
- No traps, loops, or dead-end situations were encountered during executions.

Key Finding:

- Fastest Approach: GBFS with h1
- Safest Navigation: GBFS with h2
- Most Optimal Solution: A*
- Best Trade-off for UAV Navigation: GBFS with h2 due to its balance between efficiency and risk awareness.

Complexity Analysis:

Algorithm	Time Complexity	Space Complexity	Nodes Expanded	Observation
GBFS (h1)	O(V log V)	O(V)	14	Fastest execution with minimal search effort.
GBFS (h2)	O(V log V)	O(V)	14	Slightly higher computation due to risk-aware heuristic evaluation.
A (h1)*	O(V log V)	O(V)	29	Explores more nodes to find a more optimal path.

Summary

- GBFS-h1 achieved the fastest search with the lowest computational overhead.
- GBFS-h2 provided better environmental awareness while maintaining similar search efficiency.
- A* expanded more nodes and consumed more memory but generated a lower-cost, optimal path.
- All algorithms use a priority queue, resulting in O(V log V) time complexity and O(V) space complexity.

Conclusion:

The Defence Drone Navigation System successfully applied GBFS and A* for path planning in a constrained environment. GBFS-h1 achieved the fastest navigation, while GBFS-h2 improved route selection through risk-aware decision-making. A* produced the optimal path but required higher computational resources.

GBFS-h2 offers the best balance of speed, safety, and efficiency for autonomous UAV navigation.

Plot Graph for Reference: